PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Reliability and validity of assessment methods available in primary care for bladder outlet obstruction and benign prostatic obstruction in men with lower urinary tract symptoms: a systematic review.
AUTHORS	Vredeveld, Tom; van Benten, Esther; Beekmans, Rikie E. P. M.; Koops, M. Patrick; Ket, Johannes C. F.; Mollema, Jurgen; Ramaekers, Stephan P.J.; Pool, Jan J.M.; Coppieters, Michel; Pool-Goudzwaard, Annelies L.

VERSION 1 – REVIEW

REVIEWER	Fiori, Cristian University of Turin, Oncology
REVIEW RETURNED	16-Oct-2021

GENERAL COMMENTS	Dear Authors, I read with interest your manuscript entitled “Reliability and validity of assessment methods available in primary care for bladder outlet obstruction and benign prostatic obstruction in men with lower urinary tract symptoms: a systematic review.”. I think this is a great work from a methodological point of view. I would like to congratulate the Authors approaching a wide topic like this following a rigorous methodology with COSMIN and PRISMA criteria. Here my comments. Major comments Introduction: I think is well structured and straight to the point. I would just add a paragraph on the ICS or EAU guidelines regarding the use of these measurements in clinical practice. Material and Methods The Authors should be more clear and specific regarding research strategy. The Authors stated, “All assessment methods available in primary care were considered, including questionnaires, clinical examination techniques and tests (e.g., ultrasound imaging, free uroflowmetry)”, without specifying what kind of questionnaire or clinical examination techniques were used. Please list as reported in the Results section the different kind of measurements recorded and analysed in the manuscript. This section should be structured as follows: • Search strategy and article selection: this paragraph should report inclusion (i.e. only English written manuscript) and exclusion criteria of research strategy, PICO criteria, primary and secondary outcomes • Risk of bias assessment, • Assessment of study quality, • Data extraction and analysis Table and Figure: Please add a summarizing table with current and available definition of BOO and BPO. Discussion: I will enrich the discussion with clinical implications and how the evaluation of BOO and BPO in daily urological clinical

	practice could improve patient care. Please, give some practical examples and refer to the Literature. Moreover, I think the Authors should at least mention the main therapeutic option for treatment of BOO and BPO. Minor comments • Some grammatical errors and literal fine-tuning needed. • Abbreviations to be explained on first use.

REVIEWER	Lu, Jian Peking University Third Hospital, Department of Urology
REVIEW RETURNED	24-Oct-2021

GENERAL COMMENTS	It is a well-written manuscript that systematically review the literature to determine the reliability and validity of assessment methods available in primary care to evaluate bladder outlet obstruction (BOO) and benign prostatic obstruction (BPO) in men with lower urinary tract symptoms (LUTS). The study is methodologically novel as using the COSMIN checklist for assessing the methodological quality of the included articles, which might be conducive to provide recommendations for the best assessment methods for clinicians in primary care. However, there are also several issues that need clarification and further discussion to make this paper more impactful. 1. Among the included participants in the study, are the populations with neurological disease or urethral stricture disease that might cause BOO included or excluded? 2. The description of Consensus-Based Standards for the Selection of Health Measurement Instruments (COSMIN) checklist in the part of methodological quality can be more detailed: such as how many scores can the level of evidence of study be rated as “excellent”, “good”, “fair” or “poor”?

VERSION 1 – AUTHOR RESPONSE

Reviewer comments
Reviewer 1

1. Introduction: I think is well structured and straight to the point. I would just add a paragraph on the ICS or EAU guidelines regarding the use of these measurements in clinical practice.

- Thank you for your suggestion. We have decided to change a few lines to make it clearer that we refer to the EAU guidelines (see lines 105 to 108).

1. The Authors should be more clear and specific regarding research strategy. The Authors stated, “All assessment methods available in primary care were considered, including questionnaires, clinical examination techniques and tests (e.g., ultrasound imaging, free uroflowmetry)”, without specifying what kind of questionnaire or clinical examination techniques were used.

- In this review, we included all types of questionnaires, clinical examination techniques or tests. Therefore, we specified ‘all types’ and ‘all forms’ (see line 149 to 153).

1. Please list as reported in the Results section the different kind of measurements recorded and analysed in the manuscript.

This section should be structured as follows: Search strategy and article selection: this paragraph should report inclusion (i.e. only English written manuscript) and exclusion criteria of research strategy, PICO criteria, primary and secondary outcomes Risk of bias assessment, Assessment of study quality, Data extraction and analysis Table and Figure: Please add a summarizing table with current and available definition of BOO and BPO.

- Thank you for your suggestions. We restructured the text and made several changes according to your suggestions. These include:
  - The first paragraph of the Results has been rewritten (with new subheading 'Search results and grading of evidence'), which now contains the results of the search, article selection and grading of evidence in a chronological order.
  - We added a step to the PRISMA flowchart to describe the number of articles on BOO or BPO.
  - We split Table 2 with COSMIN results into two tables. Table 3 and 4 now show the COSMIN scores per assessment method on BOO and BPO respectively, to improve readability.
  - The data extraction and analysis were provided per assessment method. We decided not to change this as we believe it provides the reader an easy and quick overview. However, we appreciated the suggestion to elaborate more on the assessment of study quality and level of evidence. Therefore, for every assessment method, we added conclusions regarding: the COSMIN scores, the consistency of findings, number of studies and total sample size to provide a better understanding of our final level of evidence grade. Accordingly, we placed the Summary of Findings table at the end of the Results section (now Table 5).
  - We added Table 1 that contains the current ICS definitions on BOO and BPO. However, we believe this would better fit the method section, and added it there (see line 139 to 140, and 144 to 146)
  - By restructuring the paragraphs, we found a small calculation error. We have accounted for this in de results section and rephrased the abstract.

1. Discussion: I will enrich the discussion with clinical implications and how the evaluation of BOO and BPO in daily urological clinical practice could improve patient care. Please, give some practical examples and refer to the Literature. Moreover, I think the Authors should at least mention the main therapeutic option for treatment of BOO and BPO

- Thank you for your comment. In the newly added second paragraph of the Discussion, entitled '2. Clinical implications and therapeutic strategies.', we expanded on the clinical implications of the evaluation of BOO and BPO. However, we believe that a paragraph on therapeutic options would be outside the scope of this review. Therefore, we mentioned therapeutic options shortly (see lines 478 to 488).

1. Minor comments: some grammatical errors and literal fine-tuning needed. Abbreviations to be explained on first use.

- Thank you, we have corrected the errors and omissions.

Reviewer 2

1. Among the included participants in the study, are the populations with neurological disease or urethral stricture disease that might cause BOO included or excluded?

- Studies with patients with neurological diseases or urethral stricture diseases as possible cause for BOO or BPO were excluded in this systematic review. We agree that this was not described clearly in the Methods and have added this in the revised manuscript (see line 156 to 159).

1. The description of Consensus-Based Standards for the Selection of Health Measurement Instruments (COSMIN) checklist in the part of methodological quality can be more detailed: such as how many scores can the level of evidence of study be rated as "excellent", "good", "fair" or "poor"?

- Thank you for your suggestion. We have made changes accordingly to provide more detail on the COSMIN description (see lines 166 to 171).

VERSION 2 – REVIEW

REVIEWER	Fiori, Cristian University of Turin, Oncology
REVIEW RETURNED	02-Feb-2022

GENERAL COMMENTS	The authors aimed to systematically review the literature regarding the reliability and validity of assessment methods for bladder outlet obstruction or benign prostatic obstruction in men with lower urinary tract symptoms (LUTS). After the analysis of about 60 papers they concluded that Transrectal or transabdominal ultrasound and uroflowmetry are the most adequate methods to evaluate BOO based on BPO in primary care and may be beneficial for an adequate referral to specialised urology care. Overall the paper is well written, and it is methodologically very well conducted. Having said that, there is nothing of new in the results of this study, US is a well establish tool for assessing BOO in men with LUTS [at least in the urological community, but also in primary care…] On the contrary results on Questionnaires are surprising in my opinion, I think that these results should be discussed more extensively. Poor level of evidence of the included papers should be stressed also in Conclusion section.

VERSION 2 – AUTHOR RESPONSE

Reviewer comments

Reviewer 1

1. Original comment: “Results on questionnaires are surprising in my opinion, I think that these results should be discussed more extensively.”

- Thank you for this comment. When considering the current EAU guideline ‘Management of Non-Neurogenic Male LUTS incl. Benign Prostatic Obstruction (BPO)’, we believe that our results on the questionnaires are in line with these guidelines. On lines 457-459, we state that the IPSS questionnaire is not able to identify BOO or BPO due to poor to moderate diagnostic accuracy. Providing additional in-depth statements on the use of this questionnaire, for example with regards to its intended use to explore symptom severity, falls outside the scope of our review.

- However, we agree that a more extensive discussion was warranted. We added a recommendation on the diagnostic accuracy of

questionnaires, which could be reviewed from a broader perspective, incorporating other measures in future research and added this to lines 542-544.

1. Original comment: “Poor level of evidence of the included papers should be stressed also in Conclusion section.”

- Based upon the advice of the reviewer we addressed the low level of evidence due to methodology and unclear definitions in the Conclusion, which is added to lines: 563-566.